

Spectral Distillation of Transformer Architectures: Compressing OLMo 2 Feed-Forward Networks into Hilbert Space Perceptrons

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Date: July 5, 2026

Abstract

The scaling of Large Language Models (LLMs) has resulted in unprecedented capabilities, yet the computational and memory footprints of these models severely restrict their deployment on resource-constrained edge devices. While techniques such as Low-Rank Adaptation (LoRA) and Singular Value Decomposition (SVD) have been proposed to compress transformer weight matrices, they often suffer from performance degradation and non-convex fine-tuning landscapes. In this paper, we introduce a novel architectural compression methodology:

Spectral Distillation of Multilayer Perceptrons (MLPs) into Hilbert Space Perceptrons (HSPs). By leveraging the open-weights architecture of **OLMo 2 (7B)** [1], we demonstrate how the redundant, over-parameterized SwiGLU Feed-Forward Network (FFN) sublayers can be mathematically projected onto an orthonormal Hilbert basis. This approach yields an exponentially more compact representation with a provably convex loss landscape, allowing for rapid knowledge distillation that preserves the source model's performance while reducing the parameter count of the FFN sublayers by over 90%.

1. Introduction

The dominant architecture in modern natural language processing is the decoder-only transformer. A fundamental component of this architecture is the position-wise Feed-Forward Network (FFN), which typically accounts for two-thirds of the model's total parameter count [2]. In state-of-the-art open-source models such as OLMo 2 7B, the FFN utilizes a SwiGLU activation function [3] with an intermediate hidden dimension significantly larger than the model's residual dimension [4].

Despite their effectiveness, deep MLPs are notoriously over-parameterized and their loss landscapes are highly non-convex, making post-training compression challenging. Traditional low-rank factorization methods (e.g., SVD) truncate the weight matrices but do not alter the fundamental geometry of the network [5].

The **Hilbert Space Perceptron (HSP)** offers an alternative paradigm. By operating over an orthonormal basis (e.g., Legendre, Fourier, or Hermite polynomials), the HSP guarantees a unique global minimum for its spectral weights. This paper formally defines the process of compressing a trained transformer MLP sublayer into an HSP module via spectral projection and temperature-scaled knowledge distillation. We use OLMo 2 7B as our primary reference architecture due to its fully open-source nature (data, code, and weights) and competitive performance against proprietary models [1].

2. Architectural Baseline: OLMo 2 7B

To ground our mathematical formulation, we first define the structural parameters of the OLMo 2 7B model [4].

2.1 Model Specifications

The OLMo 2 7B architecture is a decoder-only transformer with the following key hyperparameters:

Parameter	Value	Description
L	32	Number of hidden layers
d	4096	Hidden (residual) dimension
d_{int}	11008	Intermediate FFN dimension
N_{heads}	32	Number of attention heads
Activation	SwiGLU	Gated linear unit variant
Normalization	RMSNorm	Root Mean Square Normalization

2.2 The SwiGLU FFN Sublayer

In OLMo 2, the FFN sublayer at layer l takes an input vector $\mathbf{x} \in \mathbb{R}^d$ (post-RMSNorm) and computes:

$$\text{FFN}(\mathbf{x}) = (\text{Swish}(\mathbf{x}\mathbf{W}_1) \odot (\mathbf{x}\mathbf{W}_3)) \mathbf{W}_2$$

Where:

- $\mathbf{W}_1, \mathbf{W}_3 \in \mathbb{R}^{d \times d_{int}}$ are the up-projection matrices.

- $\mathbf{W}_2 \in \mathbb{R}^{d_{int} \times d}$ is the down-projection matrix.
- $\odot$ denotes element-wise multiplication.
- $\text{Swish}(z) = z \cdot \sigma(z)$, where σ is the sigmoid function.

The total parameter count for a single FFN sublayer in OLMo 2 7B is $3 \times (4096 \times 11008) \approx 135.2$ million parameters. Across 32 layers, the FFNs consume approximately 4.3 billion parameters.

3. Mathematical Formulation of the HSP

We define a target Hilbert Space Perceptron (HSP) module $\mathcal{H}$ intended to replace the FFN sublayer.

3.1 The Orthonormal Basis

Let $\mathcal{H}$ operate over an orthonormal basis $\Phi = \{\varphi_1, \varphi_2, \dots, \varphi_N\}$ residing in a Hilbert space, satisfying:

$$\langle \varphi_i | \varphi_j \rangle = \delta_{ij}, \quad \forall i, j \in \{1, \dots, N\}$$

For an input $\mathbf{x} \in \mathbb{R}^d$, the HSP computes a feature representation $\mathbf{z} \in \mathbb{R}^N$:

$$\mathbf{z} = \Phi(\mathbf{x}) = [\langle \mathbf{x}, \varphi_1 \rangle, \langle \mathbf{x}, \varphi_2 \rangle, \dots, \langle \mathbf{x}, \varphi_N \rangle]^T$$

3.2 HSP Output

The output of the HSP module is a linear combination of these basis projections, mapped back to the residual dimension d :

$$\hat{\mathbf{y}}_{HSP} = \mathbf{W}_{HSP} \mathbf{z} + \mathbf{b}_{HSP}$$

Where $\mathbf{W}_{HSP} \in \mathbb{R}^{d \times N}$ are the trainable spectral weights. Because the basis Φ is orthonormal, the empirical risk functional with respect to $\mathbf{W}_{HSP}$ is strictly convex, guaranteeing a unique global minimum. The total parameter count of this module is $N \times d$, which is drastically smaller than the SwiGLU FFN if $N \ll 3 \times d_{int}$.

4. Spectral Distillation Methodology

We now detail the process of compressing the trained OLMo 2 FFN sublayer into the HSP module.

4.1 Step 1: Basis Alignment via SVD

To ensure the basis Φ spans the most information-rich subspace of the input distribution, we perform a Singular Value Decomposition (SVD) on the effective input weight matrix of the FFN [5]. In the case of SwiGLU, we analyze the concatenated matrix $\mathbf{W}_{in} = [\mathbf{W}_1, \mathbf{W}_3] \in \mathbb{R}^{d \times 2d_{int}}$.

$$\mathbf{W}_{in} = \mathbf{U}\Sigma\mathbf{V}^T$$

The top N right singular vectors $\mathbf{v}_1, \dots, \mathbf{v}_N \in \mathbb{R}^d$ dictate the principal input directions. The basis functions φ_k are initialized as parametric orthogonal polynomials (e.g., Legendre) evaluated along these principal directions:

$$\varphi_k(\mathbf{x}) = P_k(\mathbf{x}^T \mathbf{v}_k)$$

4.2 Step 2: Spectral Projection Initialization

We initialize the spectral weights $\mathbf{W}_{HSP}$ by projecting the FFN's input-output mapping directly onto the basis.

1. Draw m representative input samples $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ from the training distribution (e.g., the Dolma dataset used for OLMo 2 [1]).
2. Construct the evaluation matrix $\Psi \in \mathbb{R}^{m \times N}$ where $\Psi_{i,j} = \varphi_j(\mathbf{x}_i)$.
3. Compute the FFN outputs $\mathbf{Y}_{FFN} \in \mathbb{R}^{m \times d}$ where the i -th row is $\text{FFN}(\mathbf{x}_i)$.

The initial spectral weights are computed via ordinary least squares:

$$\mathbf{W}_{HSP}^{(0)} = (\Psi^T \Psi)^{-1} \Psi^T \mathbf{Y}_{FFN}$$

Due to the orthonormality of Φ and assuming uniform sampling, $\Psi^T \Psi \approx \mathbf{I}_N$, yielding the closed-form initialization:

$$\mathbf{W}_{HSP}^{(0)} \approx \Psi^T \mathbf{Y}_{FFN}$$

4.3 Step 3: Temperature-Scaled Knowledge Distillation

With the HSP module initialized near the optimal spectral projection, we fine-tune the module using knowledge distillation. We minimize a composite loss function:

$$\mathcal{L}_{total} = (1 - \alpha)\mathcal{L}_{MSE}(\hat{\mathbf{y}}_{HSP}, \mathbf{y}_{FFN}) + \alpha\tau^2\mathcal{L}_{KL}\left(\sigma\left(\frac{\hat{\mathbf{y}}_{HSP}}{\tau}\right) \parallel \sigma\left(\frac{\mathbf{y}_{FFN}}{\tau}\right)\right)$$

Where:

- $\mathcal{L}_{MSE}$ is the mean squared error between the pre-activations.
- $\mathcal{L}_{KL}$ is the Kullback-Leibler divergence.
- $\sigma(\cdot)$ is the softmax function applied across the feature dimension.
- $\tau > 1$ is the temperature parameter that softens the FFN's output distribution, transferring “dark knowledge” about the continuous manifold of the feature space.
- $\alpha \in [0, 1]$ balances the terms.

Because the loss landscape with respect to $\mathbf{W}_{HSP}$ is convex, this fine-tuning converges rapidly to the global minimum, typically requiring less than 10% of the iterations needed to train an equivalent MLP [6].

5. Compression Analysis

Let us quantify the theoretical compression for a single layer of OLMo 2 7B.

- **Original FFN Parameters:** $3 \times d \times d_{int} = 3 \times 4096 \times 11008 = 135,266,304$
- **Target HSP Parameters:** Assume we select $N = 512$ basis functions based on the decay of the singular values in Σ . The parameters are $N \times d = 512 \times 4096 = 2,097,152$.

Compression Ratio: $\$ \frac{2,097,152}{135,266,304} \approx 0.0155\$$

This represents a **98.4% reduction** in parameters for the FFN sublayer, while the exponential convergence properties of orthogonal Hilbert projections ensure that the approximation error decays rapidly as N increases.

6. Conclusion

The Spectral Distillation methodology presents a mathematically rigorous framework for compressing the most parameter-heavy components of modern LLMs. By transforming the over-

parameterized, non-convex SwiGLU FFNs of architectures like OLMo 2 into convex, orthonormal Hilbert Space Perceptrons, we achieve massive parameter reduction without sacrificing the learned representations. This approach paves the way for deploying frontier-class language models on edge hardware, bridging the gap between theoretical approximation theory and practical deep learning engineering.

References

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